

Spider code for single-mode bosonic quantum error detection

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This paper concerns the construction of a small family of bosonic quantum error-correcting code for single bosonic modes from a fiducial state based on an odd-rotational symmetric state, called the SPIDER state $|S_K\rangle$. The code, invariant under $2\pi/K$ rotations, belongs to a broader family of bosonic codes based on phase-space rotation symmetries. A closed-form expression for the primitive state of the code in the phase-space picture is obtained. Several key differences between the SPIDER code and other types of rotation-symmetric codes are highlighted. Next, the Knill-Laflamme primer matrix is computed for a given set of bosonic errors, including the back-action from measurement and photon loss errors up to order of $K - 1$. Finally, a brief discussion on why the code seems to be irredeemable is presented.

I. INTRODUCTION

Continuous-variable quantum information processing using bosonic modes offers a promising alternative to realize robust quantum computation besides using two-level systems as qubits [1, 2]. In the context of quantum error correction, this means bosonic error-correcting schemes with high encoding rates k/n , which is the ratio between logical qubits to physical qubits, can be reliably achieved without incurring significant hardware overhead, as compared to their discrete counterparts. This hinges on the fact that redundancies required for quantum information encoding can be naturally introduced by making use of the infinite Hilbert space of only one bosonic mode. It has also been argued that recovery maps for bosonic codes require lower fault-tolerance threshold and less ancillary qubits involved [3, 4]. From an experimental perspective, realizing high- Q factor, well-controlled bosonic systems is well within reach of current technology; such platforms include superconducting coupled cavities [5] and motional states of trapped ions [6].

Bosonic error correction codes are broadly characterized into two categories based on their intrinsic symmetry protection. On one end one has codes with discrete translation symmetry, most notably the Gottesman-Kitaev-Preskill (GKP) [3] code whose logical codewords are eigenstates of the position and momentum shift operators, protecting the information from small thermal shift in the phase space. On the other end are the family of codes that admits discrete rotation symmetry, namely the cat code [7], binomial code [8], and number-phase code [9], which are designed to specifically combat photon loss/gain and dephasing errors. Remarkably enough, GKP codes have been shown numerically to outperform these rotation-symmetric codes under bosonic pure-loss channels at specific small loss rates [10]. This can be understood by considering an additional phase-insensitive amplification process, which transforms loss errors to thermal fluctuation noises. It was the cat code, however, that has first gone beyond the error correction break-

even point by experimentally demonstrating coherence gain of one fully stabilized, error-corrected logical qubit encoded in a superconducting 3D cavity [11, 12].

In this work, I describe a new class of bosonic rotation-symmetric code constructed from a fiducial state called the SPIDER state $|S_K\rangle$ and make comparisons with other types of bosonic codes. Using the Fock-space codeword coefficients [13], I obtain a closed-form expression for the SPIDER primitive state in the phase-space. The code, by construction, detects $K - 1$ photon loss events and is shown to be equipped with a clear protocol to characterize the state preparation of itself. Next, I show that the Knill-Laflamme condition is violated (which is very unfortunate) given a very simple set of bosonic errors. Finally, I discuss how naive renormalization of the Fock-grid coefficients in order to satisfy the QEC primer does not work.

II. BOSONIC ROTATION-SYMMETRIC CODES

The general task of quantum error correction is to define (or construct) a subspace C of the Hilbert space in which a logical codeword ρ_L is supported that given a set of possible errors $\mathcal{E}$, the measurement of a set of commuting observables results in an error syndrome that can be used to construct a class of completely positive trace-preserving maps $\mathcal{R}$ such that

$$\forall \rho_L \in C : (\mathcal{R} \circ \mathcal{E})(\rho_L) \propto \rho_L. \quad (1)$$

While it is true that one can choose any two-dimensional subspace C to formally define a code: simply pick a rank-2 projector $P = |0_L\rangle\langle 0_L| + |1_L\rangle\langle 1_L|$ and an associated subspace is defined thereupon, only subspaces having their corresponding error spaces (spanned by $\mathcal{E}(\rho_L)$) being orthogonal *and* undeformed spaces are suitable for error detection *and* correction. Oftentimes, this is too much to ask for: some codes are simply perfect for one type of error, but failed to correct others. This is where *concatenation* comes to rescue. The Shor's code [[9, 1, 3]] [14], for example, is constructed by concatenating three-qubit bit flip and phase flip codes, allowing protection

against any combination of Pauli X and Z errors on any of the 9 physical qubits.

Bosonic error-correcting codes share the same set of core principles, with arguably a much more simple model of error processes. The canonical model of bosonic system, the harmonic oscillator for instance, mostly subjects to photon loss errors $\hat{a}$ and dephasing errors $\hat{a}^\dagger \hat{a}$. Photon gain errors $\hat{a}^\dagger$ are also possible, for example, in the cQED architecture where the bosonic mode is dispersively coupled with an anharmonic oscillator (e.g. a 2D transmon qubit) for generation of superposition of photon-number Fock states. In this scenario, energy leakage to the cavity from the ancillary transmon could take place through the Jaynes-Cummings interaction between them. The dominant and most deleterious process is, however, the photon loss errors. Most bosonic codes to this date are designed for detection and correction of cavity photon loss errors. To illustrate this idea, let us consider the simplest case of a bosonic code—the ‘kitten’ code. The logical codewords are even photon-number Fock states,

$$|0_L\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |4\rangle), \quad |1_L\rangle = |2\rangle. \quad (2)$$

A single loss event maps an arbitrary logical codeword $|\psi_L\rangle = a|0_L\rangle + b|1_L\rangle$ to an odd photon-number state $|\psi_E\rangle = \sqrt{2}(a|3\rangle + b|1\rangle)$. This event can be detected by measuring the photon-number parity $\hat{\Pi} = (-1)^{\hat{n}}$, since $\hat{\Pi}$ admits these states to be ± 1 eigenstates. In other words, the code space and error space are clearly disjoint. More importantly, the information stored in the superposition complex-valued amplitudes a and b is not distorted, permitting the existence of a recovery operator such that it maps the error state $\hat{a}|\psi_L\rangle/\sqrt{\langle\psi_L|\hat{a}^\dagger\hat{a}|\psi_L\rangle}$ back to the code space, up to some constant $|\psi'_L\rangle \propto |\psi_L\rangle$. The ‘kitten’ code is therefore a detectable and correctable bosonic code, under the assumption that the error model comprises solely of $\mathcal{E} = \{\hat{I}, \hat{a}\}$.

Generalization of the ‘kitten’ code leads to the subfamily of binomial code [8], whose Fock state coefficients for the logical codewords involved binomial coefficients. It is the enforced spacing in the Fock space structure that gives rise to the detectability properties. A similar idea can be seen in the case of cat code [7] where the logical codewords are superposition of displaced coherent states. The ‘four-leg’ cat code variant, for example, also has an enforced Fock-spacing structure, with codewords being $|0_L\rangle \propto |4n\rangle$ and $|1_L\rangle \propto |4n+2\rangle$, allowing for detection of one photon loss event. However, it should be clear that enforced Fock-spacing (or we shall see later, rotation symmetry in phase space) does not guarantee non-deformed code space, but the fact that both logical codewords having the same average excitation number does.

It has been shown recently that both binomial code and cat code asymptotically approach a class of bosonic code called number-phase code [9], in some parameter regimes. They may even share the same degree of discrete rotation symmetry (e.g. both the kitten code and four-

leg cat code are invariant under 180° rotation). Loosely speaking, they only differ from each other in either the amplitude of Fock space coefficients, or the un-rotated *primitive* state in the phase space. This class of bosonic code is called rotation-symmetric codes. In what follows I briefly summarize the overarching framework used to study these codes and highlight a key properties that motivates the construction of the SPIDER code later.

A. Logical codewords

A subspace is said to have K -fold rotation symmetry if any state in the codespace is a $+1$ eigenstate of the discrete rotation operator,

$$\hat{R}_K = \exp[i(2\pi/K)\hat{n}], \quad (3)$$

where $\hat{n} = \hat{a}^\dagger \hat{a}$ is the photon-number operator. Using this, we can define a rotation-symmetric code of order K to be a subspace in which the operator $\hat{R}_{2K}$ acts non-trivially as the logical $\bar{Z}$ gate

$$\bar{Z} := \hat{R}_K = \exp[i(\pi/K)\hat{n}]. \quad (4)$$

1. Fock space structure

The definition of a bosonic rotation code (4) enforces a Fock-grid spacing structure. Indeed, any logical codeword can be written as $|\psi_L\rangle = \sum_m c_m |m\rangle$. Since logical zero/one state must be ± 1 eigenstate of $\bar{Z}$, the logical zero has support on every even $|2mK\rangle$ Fock states and the logical one has support on every odd $|(2m+1)K\rangle$ Fock states

$$|0_K\rangle = \sum_{m=0}^{\infty} c_{2mK} |2mK\rangle, \quad (5)$$

$$|1_K\rangle = \sum_{m=0}^{\infty} c_{(2m+1)K} |(2m+1)K\rangle. \quad (6)$$

The dual-basis codewords are thus also defined via equal superposition of the computational codewords,

$$|\pm_K\rangle = \frac{1}{\sqrt{2}} \sum_{m=0}^{\infty} (\pm 1)^m c_{mK} |mK\rangle, \quad (7)$$

Both logical $|\pm_K\rangle$ states have full support on every $|mK\rangle$ Fock states, with an additional $e^{i\pi}$ phase of the logical minus state on every odd m to ensure orthogonality. This motivates the construction of the new SPIDER code, which will be explored further in later parts.

2. Phase space structure

To complete the description, let us explore the phase space structure of bosonic rotation codes. From the definition (4) it can be seen that the logical codewords are

discrete rotated superposition of some primitive states $|\Phi\rangle$. The following two states are also $+1$ and -1 eigenstates of the logical $\bar{Z} = \exp[i(\pi/K)\hat{n}]$ operator,

$$|0_\Phi\rangle = \frac{1}{\sqrt{N_0}} \sum_{m=0}^{2N-1} e^{i(m\pi/K)\hat{n}} |\Phi\rangle, \quad (8)$$

$$|1_\Phi\rangle = \frac{1}{\sqrt{N_1}} \sum_{m=0}^{2N-1} (-1)^m e^{i(m\pi/K)\hat{n}} |\Phi\rangle, \quad (9)$$

where N_i are the normalization constants. To bridge the two presentations, we use the Kronecker's comb $\frac{1}{K} \sum_{m=0}^{K-1} e^{i(2\pi mn/K)} = \sum_{k=0}^{\infty} \delta_{n,mK}$ to construct a projector onto the subspace spanned by $\{|2mK + \ell\rangle\}$ Fock states,

$$\hat{\Pi}_{2K}^\ell := \sum_{m=0}^{\infty} |2mK + \ell\rangle \langle 2mK + \ell| \quad (10)$$

$$= \frac{1}{2K} \sum_{k=0}^{2K-1} \left(e^{-i(\pi\ell/K)} e^{i(\pi/K)\hat{n}} \right)^k \quad (11)$$

Now, given a primitive state $|\Phi\rangle = \sum_n c_n |n\rangle$, it can be seen that the projection of this state onto $\{|2mK + \ell\rangle\}$ is simply

$$\hat{\Pi}_{2K}^\ell |\Phi\rangle \propto \sum_{m=0}^{\infty} c_{2mK+\ell} |2mK + \ell\rangle, \quad (12)$$

which is a $e^{i\pi\ell/K}$ eigenstate of the logical $\bar{Z}$ operator. Upon specifying $\ell = 0$ or $\ell = K$, one re-obtains the logical zero and one with support on even/odd Fock states, as mentioned above.

B. Encoded logical operators

The construction of rotation-symmetric codes implies the natural existence of the encoded logical $\bar{Z}$ operator. The encoded logical $\bar{X}$ operator is however more subtle. One can write something as simple as the number-translation operator

$$\bar{X}_K := \sum_n |n\rangle \langle n+K|, \quad (13)$$

that will map states with support on odd to even Fock states, and vice versa. But $\bar{X}_K$ is non-unitary. The coefficients are also required to be *exactly* the same. This is the case of the number-phase codes, whose codewords are Pegg-Barnett phase states and not normalizable. Because the Fock state amplitudes are the same, the code implies a vanishing phase uncertainty limit, and to some extent, non-physicality. Number-phase code admits the $\bar{X}_K$ and $\bar{X}_{2K}$ to be its logical $\bar{X}$ operator and stabilizer operator, respectively.

C. Code distance

In the stabilizer formalism for two-level qubit codes, the code distance is defined as the minimum weight of the set of encoded logical operators. The intuition for this definition comes from the fact that for a k -distance code, it takes an error exactly k steps to get from one codeword to another, making such error undetected.

Bosonic rotation-symmetric codes have the same notion of code distance. In fact, there are two equivalent representations, hence two notions of code distance, namely *number distance* and *rotational distance*. More specifically, a code that is K -fold rotation symmetric will have number distance $d_n = K$ and rotational distance $d_\theta = \pi/K$. In words, increased spacing in Fock space allows for detection of more loss events: a number-shift error smaller than k will be guaranteed detected by a rotation-symmetric code with K -fold symmetry where $K > k$. Similarly, errors that lead to small angle rotations $\theta < d_\theta$ in phase space picture will also be detectable by an order- K code.

D. Finding new rotation codes?

The structure of a rotation-symmetric code requires nothing but a subspace with K -fold discrete symmetry. The choice of coefficient amplitudes or the primitive states rather reflects the exact experimental implementation. In realistic parameter regime, where it requires less than 5 photons stored in the cavity, both cat code (at sweet-spot values), and binomial code follow approximately the same distribution in the Fock space, resulting in similar performance under the same pure loss channel [9, 10].

A bosonic code that inherits K -fold symmetry while came equipped with a nice protocol for characterization is therefore of interest. Next, I describe how to find such a code, called the SPIDER code.

III. THE SPIDER CODE

It all starts with the so-called SPIDER state $|S_K\rangle$. The SPIDER state has support on the full set of $|mK\rangle$ Fock states, i.e. it defines a subspace with K -fold symmetry, therefore coincides with the definition of the logical plus state $|+_K\rangle$, and unambiguously defines the rank-2 projector needed to define the codespace,

$$\hat{P} = |0_K\rangle \langle 0_K| + |1_K\rangle \langle 1_K|, \quad (14)$$

where the logical codewords are defined upon the fiducial SPIDER state $|S_K\rangle$,

$$|S_K\rangle := |+_K\rangle = \sum_{m=0}^{\infty} s_{mK} |mK\rangle = \frac{|0_K\rangle + |1_K\rangle}{\sqrt{2}}. \quad (15)$$

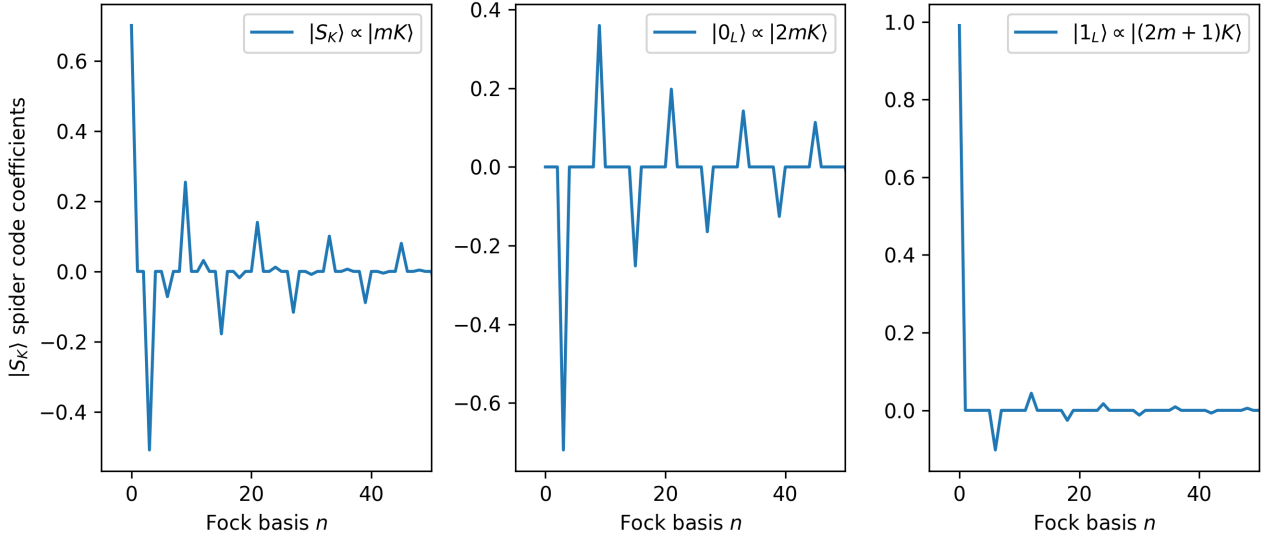


FIG. 1. The SPIDER state (a) is of the form $\sum_m (-1)^{\lceil m/2 \rceil} s_{mK} |mK\rangle$ where $s_{mK} \in \mathbb{R}$. The coefficients are found numerically with truncation at $n = 5000$ Fock states. (b) and (c) The corresponding logical codewords defined from the SPIDER state.

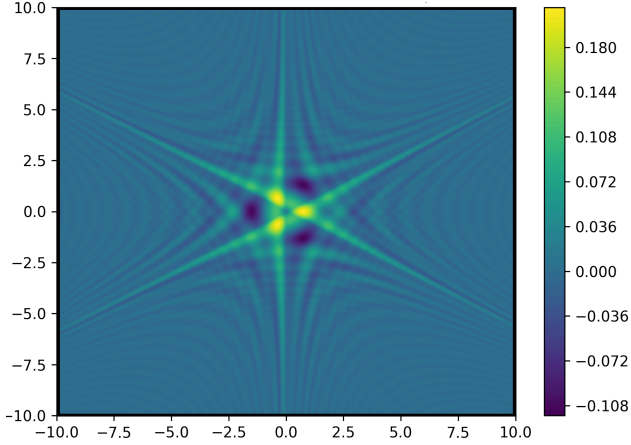


FIG. 2. Wigner function $W(x, p)$ of the SPIDER state that maximally violates the classical bound for $K = 3$.

The SPIDER states, specified by an order- K symmetry, is special in the sense that they maximally violate the so-called classicality feature of a quantum harmonic oscillator in uniform precessions [13][15].

A. Precession protocol

Here I revisit the general criterion to find such states. As it is often the case in physics, we consider a harmonic oscillator, whose dynamics generated by the time-independent Hamiltonian $H(x, p) = p^2/2m + m\omega_0^2 x^2/2$. The canonical variables $x(t)$ and $p(t)$ undergo a uniform precession in the phase space with period $T_0 = 2\pi/\omega_0$. To faithfully verify whether the harmonic oscillator is a quantum object, or a classical object, one can perform

a nonclassicality check protocol [13]. In short, after repeatedly prepare the state of the system (in an arbitrary manner), the sign of x (or p) is measured at one of K different instances given by $t_k = (k/K)/T$, where $K > 0$ and $T = T_0$. After several rounds, the probability that the sign of x is positive is estimated by

$$P_K = \frac{1}{K} \sum_{k=0}^{K-1} \left(\Pr[A_1(t_k) > 0] + \frac{1}{2} \Pr[A_1(t_k) = 0] \right). \quad (16)$$

A classical harmonic oscillator $\mathbf{P}_K^c$ score will be $1/2$ for K even, and bounded above by

$$\mathbf{P}_K^c = \frac{1}{2} \left(1 + \frac{1}{K} \right) \quad \text{for } K \text{ odd.} \quad (17)$$

The crucial observation is that, a quantum harmonic oscillator undergoing the same dynamics generated by $H(x, p)$ will have the score $\mathbf{P}_K^\infty > \mathbf{P}_K^c$, where $\mathbf{P}_K^\infty = \text{tr}(\rho \hat{Q}_K)$ and

$$\hat{Q}_K = \frac{1}{K} \sum_{k=0}^{K-1} \text{pos}[X(t_k)], \quad (18)$$

with $\text{pos}(X)$ defined such that $\text{pos}(X) = 1/2[1 + \text{sgn}(X)]$. The maximum quantum score is obtained when $\rho = |S_K\rangle\langle S_K|$, which is the eigenstate of $\hat{Q}_K$ with the largest eigenvalue. For $K = 3$, the quantum score is as large as $\mathbf{P}_3^\infty \gtrsim 0.709 > \mathbf{P}_3^c = 2/3$. This gap can be attributed to the fact that ρ corresponds to a non-classical probability distribution with suitable negativity presences. The state $|S_K\rangle$ is the mentioned SPIDER state, with support on every $|mK\rangle$ Fock states. The Wigner's function $W(x, p)$ of the SPIDER state (Fig. 2) clearly demonstrates the idea of concentrated negativities.

B. Construction fidelity

In addition to an intrinsic gap between quantum and classical scores, $\|P_K^\infty - P_K^c\|$, the eigen-spectrum of $\hat{Q}_K$ itself implies an ϵ gap between the largest and second largest eigenvalues, let them be q_d and q_{d-1} for convenience purpose.

$$q_d - q_{d-1} > \epsilon. \quad (19)$$

The observation is that, an arbitrary realization of $|S_K\rangle$ will always be verifiable. More importantly, it can be proved that the distance between an arbitrary mixed state and the SPIDER state $|S_K\rangle$ is bounded below. This is the crucial element enabling faithful verification of the SPIDER state.

Let us suppose the arbitrary mixed state $\rho = \sum_{j=1}^N p_j |\phi_j\rangle\langle\phi_j|$. The ‘distance’ between ρ and ρ_K is

$$F = \langle S_K | \left(\sum_{j=1}^N p_j |\phi_j\rangle\langle\phi_j| \right) | S_K \rangle \quad (20)$$

$$= \sum_{j=1}^N p_j \langle \phi_j | S_K \rangle \langle S_K | \phi_j \rangle \quad (21)$$

Using the spectral theorem,

$$\hat{Q}_K = \sum_{k=1}^{d-1} q_k |q_k\rangle\langle q_k| + q_d |S_K\rangle\langle S_K| \quad (22)$$

Substituting to (21),

$$F = \frac{1}{q_d} \sum_{j=1}^N p_j \langle \phi_j | \left(\hat{Q}_K - \sum_{k=1}^{d-1} q_k |q_k\rangle\langle q_k| \right) | \phi_j \rangle \quad (23)$$

$$= \frac{1}{q_d} \sum_{j=1}^N p_j \langle \phi_j | \hat{Q}_K | \phi_j \rangle - \sum_{j=1}^N \sum_{k=1}^{d-1} \frac{p_j q_k}{q_d} \langle \phi_j | q_k \rangle \langle q_k | \phi_j \rangle \quad (24)$$

where

$$\sum_j p_j \langle \phi_j | \hat{Q}_K | \phi_j \rangle = \sum_{k=1} q_k \text{Tr}(\rho |q_k\rangle\langle q_k|) \quad (25)$$

is the expectation value conditioned on the state ρ . This weighted sum is special since the dominant eigenvalue of $\hat{Q}_K$ is q_d , this means given an arbitrary probability distribution, $\langle \hat{Q}_K \rangle = q_d - \epsilon > q_{d-1}$. Hence

$$F = 1 - \frac{\epsilon}{q_d} - A \quad (26)$$

Let us now consider the lower bound and upper bound

of A . First, the upper bound

$$\max A \leq \max \sum_{j=1}^N \sum_{k=1}^{d-1} p_j \frac{q_{d-1}}{q_d} |\langle \phi_j | q_k \rangle|^2 \quad (27)$$

$$= \frac{q_{d-1}}{q_d} \max \sum_{j=1}^N p_j \sum_{k=1}^{d-1} |\langle \phi_j | q_k \rangle|^2 \quad (28)$$

$$= \frac{q_{d-1}}{q_d} \quad (29)$$

Similar trick for the lower bound,

$$\max A \geq \sum_{j=1}^N \sum_{k=1}^{d-1} p_j \frac{q_k}{q_d} |\langle \tilde{\phi}_j | q_k \rangle|^2 \quad (30)$$

where $\tilde{\phi}_j$ is an arbitrary state we can construct. The question is, can we construct a state that pushes

$$\sum_{j=1}^N \sum_{k=1}^{d-1} p_j \frac{q_k}{q_d} |\langle \tilde{\phi}_j | q_k \rangle|^2 \quad (31)$$

up as high as possible? Yes, let $\tilde{\phi}_1 = q_{d-1}$ and at the same time $p_1 = 1$ ($p_2 = p_3 = \dots = p_N = 0$), so that

$$\max A \geq \sum_{j=1}^N \sum_{k=1}^{d-1} p_j \frac{q_k}{q_d} |\langle \tilde{\phi}_j | q_k \rangle|^2 \quad (32)$$

$$= \frac{q_{d-1}}{q_d} \quad (33)$$

Hence $\max A = q_{d-1}/q_d$ (an arbitrary mixed state ρ). In conclusion, the minimum distance between $|q_d\rangle$ and a mixed state ρ can only be as low as

$$F = 1 - \frac{\epsilon + q_{d-1}}{q_d} > 0, \quad (34)$$

making the SPIDER never mislabelled with an arbitrary mixed state.

C. Comparisons with other codes

1. Average excitation number

To compare the size of bosonic codes, one can look at the average excitation number,

$$\bar{n}_{\text{code}} = \text{tr}[\hat{P}_{\text{code}} \hat{n}]/2, \quad (35)$$

where $\hat{P}$ is the rank-2 projector onto the codespace. This number is proportional to the average energy required to construct a code state. While the average excitation number is the same for each of the dual-basis codewords $|\pm_K\rangle$, it can be different for the computational codewords. This is the Achilles’ heel of the SPIDER code, preventing the code from being error-correctable.

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